Monic remainder loop: successful empty return implies divisibility

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The remainder model selects offset len(r)-len(b) and multiplier last(r), matching native pdiv's monic-divisor branch. It runs the previously checked word-array cancellation and trims the result. Canonical lists remain canonical through each step. The outer loop stops on an empty remainder or one shorter than the divisor. Each successful returned remainder has exactly the same divisibility-by-b predicate as the initial polynomial.

A checked wrapper requires the divisor's last word to be one, which implies a nonempty divisor. If this wrapper returns Some(empty), the divisor polynomial divides the input polynomial. The initialization trim is included in the proof. The result composes word arithmetic, bounded indexed updates, generated divisor events, polynomial cancellation, trimming and recursive invariant preservation. The quotient array is not modeled because prem returns only the remainder.

Fuel counts allowed cancellation steps. Exhaustion returns None, distinct from Some(empty). Terminal inputs are accepted even with zero fuel. The proof establishes soundness of successful returns for any fuel, not that a particular fuel always suffices or that the native unbounded loop terminates. A decreasing-length/degree bound remains the next proof obligation. The checked wrapper's rejection behavior is a proof-harness interface; no production Rust behavior was changed.

Nine main declarations and ten controls are exposed for separate axiom readback. Modulo17, division of X^2-1 by X-1 returns Some(empty) with two steps, while one step returns None. X^2+1 returns Some([2]). Controls also cover zero fuel, already-short remainder, initial trailing zeros, empty and nonmonic divisor rejection, zero input and division by constant one. These are kernel-evaluated controls, not measured runs.

Assumptions: prime $0<p<2^{31}$ and canonical divisor/input words. Native interpretation is the odd prime-field branch with diagnostic tables disabled and monic divisor; counters and quotient writes are outside this model. The separate theorem proving the top-one and sliced-array validity guards supplies the intended native shape. A formal equivalence to compiled Rust and an independent source-fidelity verdict remain unestablished.

Sources: native three_leads.rs pdiv/prem, prior trimmed_polynomial_step/run_correct, pinned List get-Last/membership lemmas and divisibility subtraction. No added compiled dependencies. Five final native checks passed with unchanged input bindings. All 19 separate readbacks use only standard axioms; results are in status.json. Initial outer-loop proof over-counted the impossible fuel-exhaustion branch already eliminated by simplification; corrected proof and failed attempt are retained. Initial controls could not reduce the well-founded recursion wrapper; explicit structural recursion on fuel and rebuilt dependents passed kernel reduction. Existing compiled dependencies retain their trust boundary. No scientific execution or cost ratio.